


# Mateusz Kaczmarek

Hardware / PCB Engineer


- Student status (Active until July 2027)
- Working hours: Full-time (Available immediately)
- Work model: Hybrid preferred (On-site possible)
- Active category B driving license

## Tools

| | |
|---|---|
| Altium Designer | Mid/Adv. |
| LTspice / LTPowerCAD / QUCS | Mid/Adv. |
| SI & PI / HyperLynx | Intermediate |
| HFSS / FEKO / AWR | Beginner |
| HW / Mechanical Integration | Intermediate |
| CAD (Inventor, Fusion, DesignSpark) | Intermediate |
| DSP (STM32 / FPGA) | Beg./Mid. |
| C / C++ / Python / MATLAB | Beg./Mid. |
| Linux Embedded | Beginner |
| Git / LaTeX | Intermediate |

Measuring equipment:
VNA, oscilloscopes, spectrum analyzers, logic analyzers, arbitrary waveform generators, multimeters.

## Languages

| | |
|---|---|
| Polish | Native |
| English | B2 |

## Profile

Electronics engineer specializing in the design of advanced printed circuit boards (High-Speed PCB), RF circuits, power electronics, as well as embedded, analog, and mixed-signal systems. I have a rich, interdisciplinary portfolio that goes well beyond academic standards. I love creating devices from A to Z – from concept, through schematics, simulations, SI/PI analysis, and PCB layout, to mechanical integration, firmware development, and final hardware bring-up.

## Key Competencies

- Analog and Digital Electronics (Mixed-Signal): Design of complex analog, digital, and mixed-signal circuits based on technical documentation, datasheets, and application notes. Development of sensitive, low-noise analog paths (e.g., precise audio paths) and design of advanced digital systems utilizing microcontrollers, SoCs, and FPGAs.
- Power Electronics and Power Supply: Design of DC-DC converters, LDO regulators, PoE++ systems, and advanced isolated topologies (e.g., 500W LLC with PFC). Power Integrity (domain division and organization), precise Power Sequencing for SoC and FPGA systems, and Thermal Management of high-power components.
- High-Speed PCB & SI: Over 8 years of continuous development in PCB design, including the last 2 years of intensive work on advanced multi-layer boards (up to 8 layers) as part of degree projects. Practical knowledge of routing rules, impedance control, and length tuning for interfaces such as LPDDR4, DDR3, PCIe, QSGMII, Ethernet, QSPI, OctoSPI, eMMC, HDMI, USB, and DP.
- RF and Lab Measurements: Theoretical and practical foundations of microwave electronics. Experience in simulations (LTspice, QUCS, HFSS), matching circuit design, and testing basic antenna characteristics (e.g., patch, dipole) for Wi-Fi and Bluetooth systems. Solid proficiency in operating laboratory equipment for the bring-up and debugging of mixed-signal circuits.
- ECAD-MCAD Integration, Prototyping, and Assembly: Complex 3D modeling (Autodesk Inventor, DesignSpark), FDM/SLA 3D printing, and mechanical parts design (e.g., enclosures, heatsinks). Generating production documentation (BOM, Gerbers), coupling mechanics with electronics, EMI shielding, and component assembly (SMD/THT, soldering irons, Hot-Air).

## Experience and Projects

Lead Designer (M.Sc. Thesis) — 10/2025 - Present
Universal Modular Measurement System

Hardware architecture design of an advanced measurement system. Design work (schematics and advanced PCB layout) has so far included: an 8-layer SoM compute module with a Rockchip RK3588 processor (LPDDR4 and PCIe routing with impedance control and length-tuning), a 6-layer communication node based on the LAN9696 TSN switch (hardware PTP IEEE 1588v2 support, DDR4, QSGMII), and a multi-section power supply system. A 500W AC/DC resonant LLC half-bridge converter with active PFC and a dedicated PoE++ controller (LTC4291/LTC4292) with full galvanic isolation were designed.

Systems Engineer — 02/2026 - 06/2026
Adaptive Sonar System (SRMP)

Design of an ultrasonic sonar (STM32) with advanced stochastic data fusion (IMM-AEKF). Integrated a custom linear array of transducers, utilized hardware DWT counters for precise time measurement, and implemented a feedback loop in the MATLAB environment. Achieved millimeter-level precision in tracking dynamic objects.

## Soft Skills

- Communication and teamwork
- Independence and initiative
- Strong drive for learning
- Meticulousness and dedication

## Education

- 10/2025 - Present

Master of Science in Engineering (M.Sc. Eng.)
Warsaw University of Technology (WEiTI)
Electronics, specialization: Electronic and Embedded Systems.

- 10/2021 - 09/2025

Bachelor of Science in Eng. (B.Sc. Eng.)
Warsaw University of Technology (WEiTI)
Electronics, specialization: Electronics and Photonics
B.Sc. Thesis: "Advanced Audio-Visual System".

- 09/2017 - 05/2021

Technician
Technical School No. 1 in Piaseczno
Electronic profile.

## Interests

- 3D printing, amateur robotics & modeling
- Hi-Fi audio, Polish rock, vinyl records
- Sport shooting, archery
- LEGO models
- Aquascaping
- Playing guitar

## Contact


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github.com/MKaczmarek-HW


Firmware and GUI Developer — 02/2026 - 06/2026
Data Acquisition System (STM32 + Raspberry Pi)

Development of a mini-oscilloscope and spectrum analyzer in a Master-Slave architecture. Implemented firmware in C (STM32), hardware acquisition triggering (Analog Watchdog), and transfer optimization via DMA/SPI. Designed a GUI application in PyQt5 for real-time logarithmic FFT analysis.

Digital Logic Designer — 10/2025 - 02/2026
VGA Video Controller SoC (FPGA Zynq-7000)

Implementation of a hardware SoC system on the Zybo Z7-10 platform. Integrated a custom VGA controller (VHDL) with an ARM Cortex-A9 processor via the AXI4 bus. Deployed bare-metal drivers in C, dynamic clock reconfiguration (DRP), pipelined image generators (4-stage delay line), and clock domain crossing (CDC) management.

Lead Designer (B.Sc. Thesis) — 10/2024 - 09/2025
Advanced Audio-Visual System

Independent design of a Mixed-Signal system integrating low-noise audio paths (MM/MC) with a digital control unit (STM32H7). Implemented an HMI interface (TouchGFX, 24-bit LTDC) with contactless magnetic encoders and a multi-section linear power supply (4 transformers, Soft-Start). Full cycle: from ECAD/MCAD design to physical hardware bring-up.

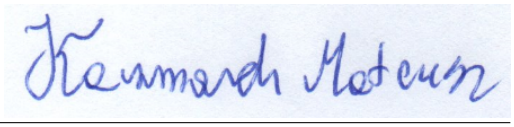
Electronics Engineering Intern (Hardware & Firmware) — 07/2024
Etronika

One-month internship in the R&D department. Main responsibilities included printed circuit board (PCB) design and programming STM32 microcontrollers in C. Gained practical understanding of the commercial electronic device design and implementation cycle.

Lead Designer — 2018 - 2021
Mechatronic Projects: Scorpion and Robotic Arm

Complex systems combining 3D CAD modeling (FDM printing) with Arduino Mega microcontroller programming. The 'Scorpion' robot integrated 35 independent degrees of freedom controlled by an I2C bus (PCA9685 cascade) and a 5.8GHz FPV video link. The robotic arm was based on a NEMA17 stepper motor and servos, featuring a fully functional HMI panel with an E-Stop switch.

I consent to the processing of my personal data for the purposes of the recruitment process (in accordance with the Act of May 10, 2018, on the protection of personal data (Journal of Laws of 2018, item 1000) and pursuant to the Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (GDPR)).



Warka, November 7, 2026